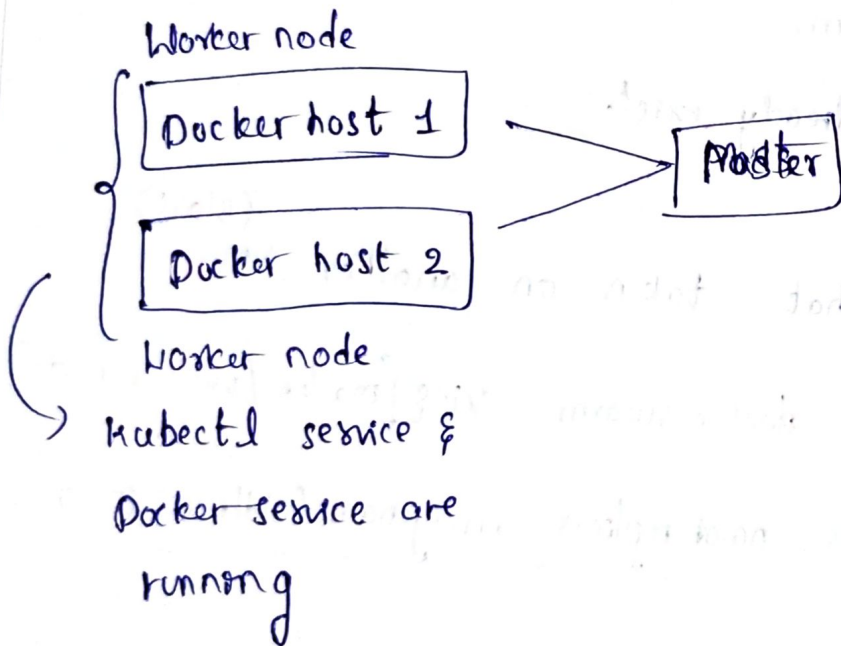


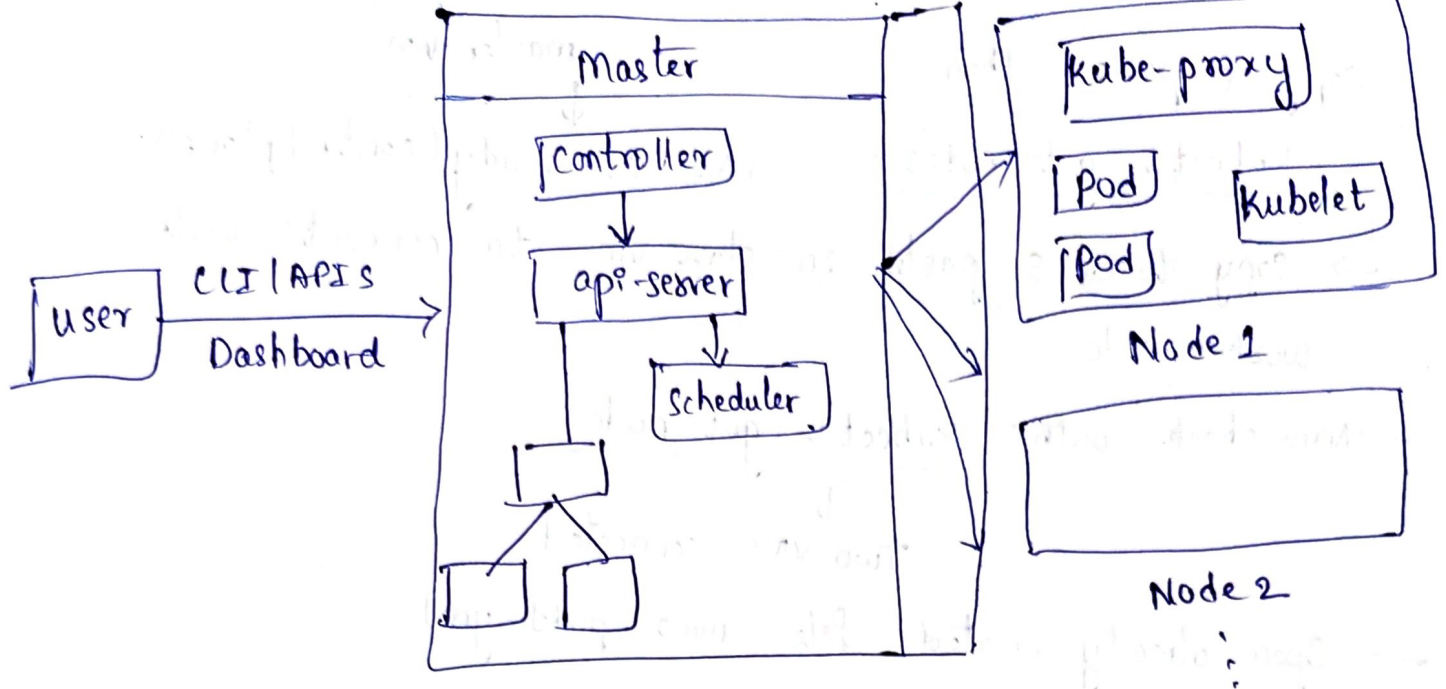
KUBERNETES

- To manage the n/w traffic/load to share containers to distributed parties/ Nodes.
- Kubernetes is an orchestration framework for Docker container which helps expose containers as services to the outside world.
- Deploy containerized application on cluster
- Scale deployment
- Update containerised app with new software version
- Debug the containerised application

In docker container = kubernetes pod



clustering — manages nodes, pods etc



- kube-proxy - Helps to connect with network
- kubelet - Helps to connect with master.
- Automated pods are scheduled by scheduler
- controller - controls all the pods.
- If we want to deploy pods in global level we use kubernetes.

Pod Architecture

- Pod automatically assign dynamic Ip address.

Steps in kubernetes

- kubectl get nodes [Error] [In master node]
- To remove kubernetes - kubeadm reset
- To install kubernetes - kubeadm init

There we get commands

```

mkdir -p $HOME/.kube
sudo cp -r /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u) : $(id -g) $HOME/.kube/config

```

copy & paste then

→ kubectl get nodes

master VM
↓
ready (control plane)
- VM1 is ready (control plane)
to connect with

→ copy token & paste in slave VM
worker node

→ Now check with kubectl get nodes

↓
Two VMs connected

→ Open already created - file nano pod1.yaml.
change yaml code

→ To run container/pod

→ kubectl apply -f pod1.yaml → ~~new~~ pod will be created

→ kubectl get pods // container gets created.

if container is still creating. Then we have to do.

[Kubernetes cluster ubuntu 22.04] To get all Kubernetes steps.
↳ Install Calico Network plugin - To enable communication
between pods. [copy & paste cmd]

→ Now verify it properly runs or not

kubectl get pods -n kube-system

kubeg. kubectl get pods

Then pod status is changed to running

→ kubectl execute podname

↳ kubectl exec podname -it /bin/bash To get

inside the pod.
@root → @podname

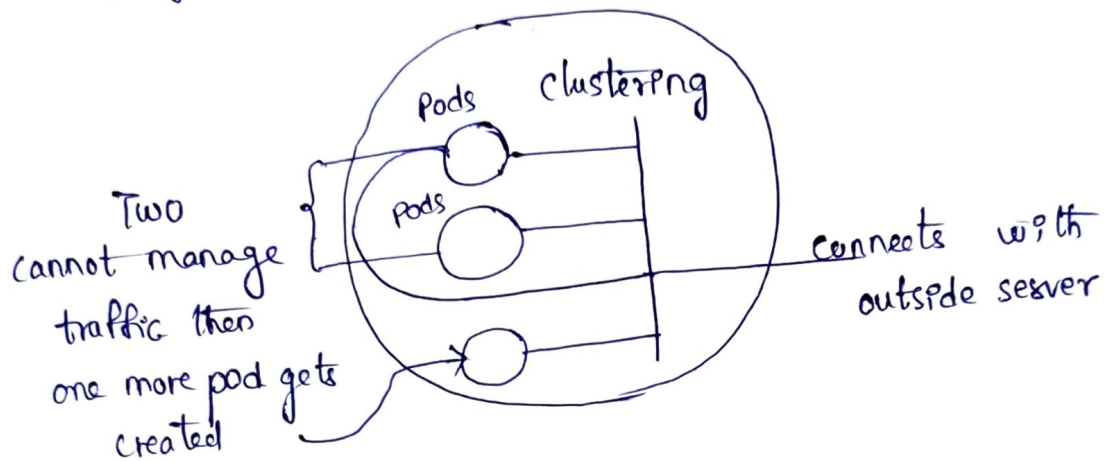
→ We get pod ip address with ifconfig

→ apt install apache2 | get or any

in another terminal all on root only.

→ kubectl get all -o wide [shows all running pods with status]

12/10/24 Kubernetes deployment



→ Creating deployment file

nano deploy.yaml (or) nano deployment.yaml
[Copy code from browser & paste]

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
```

```
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
```

```
spec:
  containers:
    - name: nginx
      image: nginx:1.1:2
      ports:
        - containerPort: 80
```

(apiVersion:)

→ kubectl apply -f deployment.yaml To create pod

→ To list all pods kubectl get pods

→ To delete pod kubectl delete pod podname